

Installation Guide

RGB Status LED (WS2812B) + OLED Display (SSD1306) for Raspberry Pi 4 · `status_led.py`

This guide sets up a status LED (WS2812B) and, in parallel, a 128×32 OLED (SSD1306 / Adafruit PiOLED) on a Raspberry Pi 4. The LED shows the system state by colour, the OLED shows the same in plain text (IP, CPU, RAM, status). Both are driven by a single script and one service. The last section is a hands-on “Troubleshooting” chapter with the pitfalls from a real-world setup.

What the LED shows

- **LED green** — normal operation; bright flicker on disk activity
- **LED red/green alternating every second** — over-temperature (threshold configurable)
- **LED blue blinking** — no network connection
- **LED magenta blinking** — backup failed
- **LED cyan pulsing** — backup running
- **OLED** — continuously shows IP address, CPU temperature + load, RAM usage and the state in plain text

1. Requirements

Hardware

- Raspberry Pi 4 with Raspberry Pi OS (Bookworm) — **with a physical GPIO header**
- WS2812B LED (5050), single or as a strip
- OLED display 128×32, SSD1306, I2C (e.g. Adafruit PiOLED)
- Resistor 330–470 Ω for the LED data line
- Jumper wires; for multiple LEDs also a level shifter (74AHCT125) and a 1000 µF capacitor

Software

- Python 3 (pre-installed on Raspberry Pi OS)
- Internet access for the one-time library installation
- The file `status_led.py` (from this repository)

2. LED connection method: PWM or SPI

The WS2812B can be driven two ways. The OLED is unaffected (separate bus).

- **PWM (GPIO18) — recommended**, especially for a single LED. The data line is driven cleanly and pulled Low when idle — “off” stays off and weak colours (dim green) stay stable. Requirement: the service runs as **root** and the onboard audio must be disabled.
- **SPI (GPIO10/MOSI)** — runs without root and without the audio conflict. **However:** the Pi leaves the SPI data line High between packets; some WS2812B read this as “all bits on” = white. Dim colours and “off” can then drift to white (see Troubleshooting). Only use this if root/audio are a problem and the LED passes the test.

Tip: When in doubt, use PWM — it is the robust default for a single WS2812B on the Pi. This guide is written for PWM; the SPI steps are noted as an alternative throughout.

3. Wiring (LED + OLED)

Power off the Pi and connect both. The pins of the two devices do not overlap:

WS2812B (LED)

- **DIN** (mind the arrow — input side) → through the **330-470 Ω resistor** to **GPIO18 / pin 12** (PWM) or GPIO10/MOSI (SPI)
- **VDD** → for a **single** LED to **3.3 V (pin 1)**; for multiple LEDs to 5 V *with* a level shifter
- **GND** → GND of the Pi (a common ground is mandatory)

Tip: VDD at 3.3 V instead of 5 V is the simplest and clean permanent solution for a single LED: the Pi outputs data at 3.3 V, but a WS2812B at 5 V expects ~3.5 V for “High” — at 3.3 V VDD the level matches again. For multiple LEDs (5 V required) a level shifter (74AHCT125) is needed.

OLED (SSD1306, I2C)

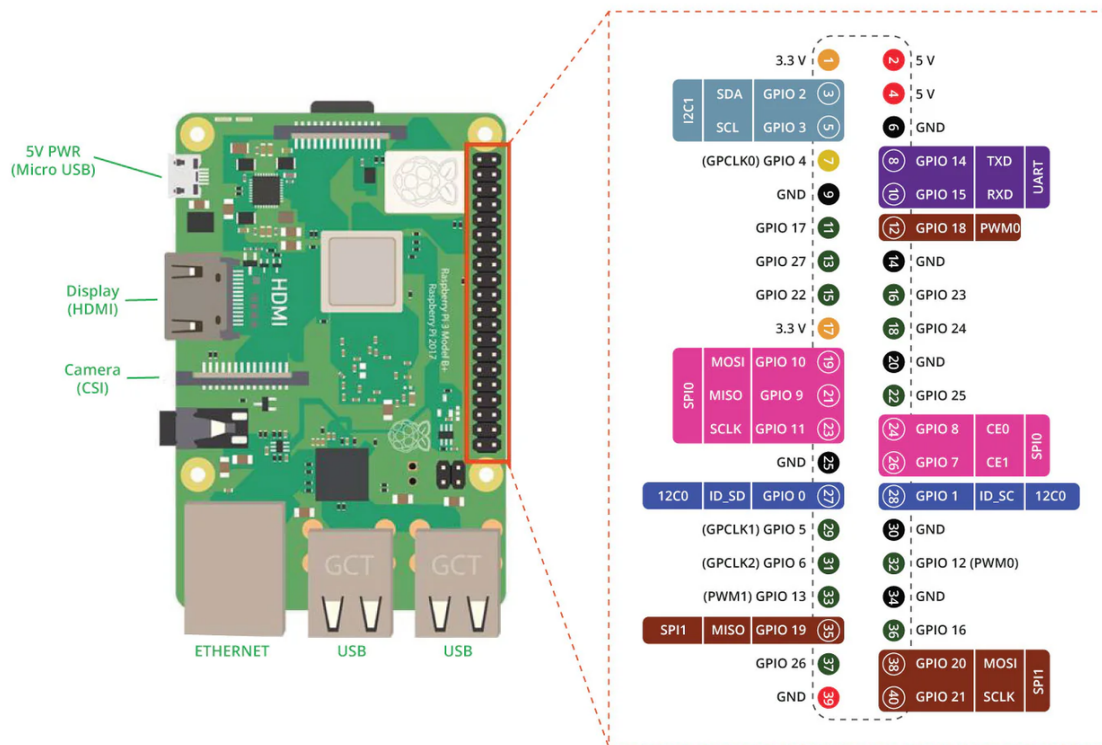
- **3V3** → pin 1 · **GND** → GND
- **SDA** → GPIO2 (pin 3) · **SCL** → GPIO3 (pin 5)

Combined pin overview

Device / function	Pin	Signal
LED – data (PWM, rec.)	Pin 12	GPIO18
LED – data (SPI, alt.)	Pin 19	GPIO10 / MOSI
LED – VDD (1 LED)	Pin 1	3V3
LED – GND	Pin 9 (free)	GND
OLED – 3V3	Pin 1	3V3
OLED – SDA	Pin 3	GPIO2
OLED – SCL	Pin 5	GPIO3
OLED – GND	Pin 6	GND

Pin 1 (3V3) powers both the OLED and the single LED — both may share it. A PiOLED usually sits on the corner pins (1/3/5 + GND), so take the LED's GND from a free pin (e.g. pin 9).

Raspberry Pi 4 - pinout (reference)



Assignment for this project: LED data to GPIO18 (pin 12), LED VDD and OLED 3V3 to 3.3 V (pin 1), OLED SDA to GPIO2 (pin 3), OLED SCL to GPIO3 (pin 5), common ground to a GND pin (e.g. pin 6 or pin 9). SPI alternative for the LED data: GPIO10/MOSI (pin 19).

4. Do OLED and LED work together?

Yes. The OLED is on the **I2C bus** (GPIO2/3), the LED on the **PWM line** (GPIO18) or the **SPI bus** (GPIO10). Separate hardware buses on different pins — no conflict. Both are driven by the same process (`status_led.py` refreshes the OLED once per second). If the OLED fails to initialise, the LED still runs, and vice versa.

Requirement: real GPIO/I2C/SPI interfaces of a Raspberry Pi. In an LXC container or a VM (e.g. on an x86 Proxmox host) there is no Pi header — then none of this works. Quick check whether it is a real Pi with both buses:

```
cat /proc/device-tree/model      # must show 'Raspberry Pi 4 ...'
ls -l /dev/i2c-1 /dev/spidev0.0 # both device files must exist
```

5. Install the script

Copy the file into a fixed location and make it executable:

```
sudo cp status_led.py /usr/local/bin/status_led.py
sudo chmod +x /usr/local/bin/status_led.py
```

6. Install the libraries

Each command is deliberately on **one line** — a backslash at the end of a line is not needed (and, in the middle of a line, causes errors).

PWM (recommended)

```
sudo apt install -y python3-pil i2c-tools
sudo pip3 install adafruit-blinka --break-system-packages
sudo pip3 install rpi_ws281x --break-system-packages
sudo pip3 install adafruit-circuitpython-neopixel --break-system-packages
sudo pip3 install adafruit-circuitpython-ssd1306 --break-system-packages
```

SPI (alternative)

```
sudo apt install -y python3-pil i2c-tools
sudo pip3 install adafruit-blinka --break-system-packages
sudo pip3 install adafruit-circuitpython-neopixel-spi --break-system-packages
sudo pip3 install adafruit-circuitpython-ssd1306 --break-system-packages
```

Alternative — virtual environment (without touching the system Python)

```
sudo apt install -y python3-venv i2c-tools
sudo python3 -m venv /opt/status-led-venv
sudo /opt/status-led-venv/bin/pip install adafruit-blinka rpi_ws281x
sudo /opt/status-led-venv/bin/pip install adafruit-circuitpython-neopixel
sudo /opt/status-led-venv/bin/pip install adafruit-circuitpython-ssd1306 pillow
```

With the venv variant, use the interpreter from the venv in the service (section 10):
ExecStart=/opt/status-led-venv/bin/python /usr/local/bin/status_led.py

7. Enable I2C, check the OLED, (PWM:) disable audio

Enable I2C for the OLED (for the SPI LED also enable SPI):

```
sudo raspi-config
# Interface Options -> I2C -> Yes
# (SPI variant only: Interface Options -> SPI -> Yes)
```

For the **PWM variant** the onboard audio must be off (it shares the PWM hardware, otherwise the LED flickers). In `/boot/firmware/config.txt`:

```
sudo sed -i 's/^dtparam=audio=on/dtparam=audio=off/' /boot/firmware/config.txt
grep audio /boot/firmware/config.txt
# if grep shows no 'audio=off', then once:
echo "dtparam=audio=off" | sudo tee -a /boot/firmware/config.txt
```

After the interface changes, reboot and look for the OLED on the I2C bus:

```
sudo reboot
# after the reboot:
sudo i2cdetect -y 1
```

If '3c' appears in the table, the display is detected. If yours shows 0x3d, enter that value in the config at `oled_addr`.

8. Adjust the configuration

Open the configuration block at the top of `status_led.py` (excerpt):

```
sudo nano /usr/local/bin/status_led.py

@dataclass
class Config:
    led_type: str = "ws2812"          # PWM (recommended); SPI: "ws2812-spi"
    ws_count: int = 1                 # the actual number of LEDs!
    ws_brightness: float = 0.50       # master brightness 0..1
    ws_pixel_order: str = "GRB"       # standard WS2812B; rarely "RGB"
    oled_enabled: bool = True         # drive the OLED in parallel
    oled_addr: int = 0x3C              # I2C address (see i2cdetect)
    temp_threshold_c: float = 70.0    # over-temperature threshold in deg C
    net_check_host: str = "1.1.1.1"  # internet check; for LAN use gateway IP
```

- **led_type** — "ws2812" (PWM) or "ws2812-spi" (SPI)
- **ws_count** — **exactly** the number of your LEDs; too low → surplus LEDs keep their old value
- **ws_pixel_order** — almost always "GRB"; only change it if the colour test says so
- **ws_brightness** — master dimmer; WS2812B are very bright, 0.2–0.5 is usually pleasant

9. Functional test

Without hardware (terminal)

Shows the LED states and the OLED lines in parallel in the terminal:

```
python3 status_led.py --simulate --duration 30
```

Direct LED test (colours + stability, PWM)

Checks the colour order and whether dim green stays stable (with `sudo`, GPIO18):

```
sudo systemctl stop status-led.service
sudo python3 - <<'PYEOF'
import board, neopixel, time
px = neopixel.NeoPixel(board.D18, 1, brightness=0.3, auto_write=True, pixel_order="GRB")
for n,c in [("RED", (255,0,0)), ("GREEN", (0,255,0)), ("BLUE", (0,0,255))]:
    px.fill(c); print("commanded:", n); time.sleep(2)
print("20s dim green - stays green, then off cleanly?")
for i in range(200): px.fill((0,40,0)); time.sleep(0.1)
px.fill((0,0,0))
PYEOF
```

Red/green/blue must match, the dim green must stay stable and turn off cleanly at the end. For the SPI variant use `neopixel_spi.NeoPixel_SPI(board.SPI(), 1, ...)` instead.

Service in the foreground (shows errors live)

```
python3 /usr/local/bin/status_led.py    # stop with Ctrl+C
```

10. Autostart as a service (systemd)

One service drives the LED and the OLED together. Create the service file (template in this repo):

```
sudo nano /etc/systemd/system/status-led.service
```

```
[Unit]
Description=RGB Status LED + OLED
After=network-online.target
Wants=network-online.target

[Service]
Type=simple
ExecStart=/usr/bin/python3 /usr/local/bin/status_led.py
Restart=always
RestartSec=5
User=root
RuntimeDirectory=status-led
# SPI variant: User=YOUR_USERNAME (NOT 'pi')
# venv: ExecStart=/opt/status-led-venv/bin/python /usr/local/bin/status_led.py

[Install]
WantedBy=multi-user.target
```

Tip: Important about the user: **PWM needs User=root**. For the SPI variant enter a real username — current Raspberry Pi OS no longer has the former default user pi. A wrong/missing user leads to status=217/USER (see Troubleshooting).

Enable, start and check:

```
sudo systemctl daemon-reload
sudo systemctl enable --now status-led.service
systemctl status status-led.service
journalctl -u status-led -f
```

For the SPI variant as a normal user, that user must be in the groups 'spi' and 'i2c' (default on Raspberry Pi OS): `groups YOUR_USER` and `sudo usermod -aG spi,i2c YOUR_USER`.

11. Connect a backup status (optional)

The script reads the backup state from `/run/status-led/backup`. If the file is missing, no backup state is shown. Your backup job writes running, ok or failed into it:

```
#!/bin/bash
mkdir -p /run/status-led
echo running > /run/status-led/backup

if restic backup /data; then
    echo ok > /run/status-led/backup
else
    echo failed > /run/status-led/backup
fi
```

The failure state (LED magenta, OLED “Backup-Fehler!”) stays until the file is set back to 'ok'.

12. Reference: states, LED colour and OLED text

State	LED colour & pattern	OLED text	Prio
Over-temperature	Red/green, 1 Hz	UEBERTEMPERATUR!	100
No network	Blue, 2 Hz blinking	Kein Netzwerk	80
Backup failed	Magenta, 2 Hz	Backup-Fehler!	70
Backup running	Cyan, pulsing	Backup laeuft...	40
Normal operation	Green (bright on I/O)	Normalbetrieb	0

When several states are active, the one with the highest priority wins (for both LED colour and OLED text). The OLED also permanently shows IP, CPU temperature + 1-minute load and RAM. The OLED texts are defined in STATUS_TEXT (German by default) and can be localised there.

13. Add your own states

Each state consists of a condition and a render function and is registered with a priority in the STATUSES list. For the OLED display, also add a text in STATUS_TEXT:

```
def is_my_state(ctx):
    return ...                # True when active

def render_my_state(ctx):
    return (1.0, 0.5, 0.0)    # colour (R, G, B), each 0..1

STATUSES.append(StatusDef("my_state", priority=50,
                          condition=is_my_state,
                          render=render_my_state))

STATUS_TEXT["my_state"] = "My text"
```

14. Troubleshooting (from practice)

The following cases come from a real setup — from the service start to the flickering LED.

► Service does not start, status “status=217/USER” in the log

Cause: The user given in the .service file does not exist. Current Raspberry Pi OS no longer has the former default user pi.

Fix: For PWM set User=root; for SPI use the real username. Then daemon-reload and restart.

► pip error “externally-managed-environment”

Cause: On Bookworm the system Python is protected (PEP 668) — or a backslash sits in the middle of the line instead of at the end and breaks the command.

Fix: Put each command on **one** line and add --break-system-packages at the end, or use the venv variant (section 6).

► LED does not light at all, although the script runs without errors

Cause: Wiring (DIN/DOUT swapped, no common ground) or a logic level that is too low (VDD at 5 V, data only 3.3 V).

Fix: First check full brightness with the direct test (section 9). If nothing lights: check the DIN side and GND, then move VDD from 5 V to **3.3 V** as a test. For a single LED, 3.3 V is the permanent solution.

► **Wrong colours (e.g. red commanded, green lights up)**

Cause: The colour-channel order of the LED differs. The standard is GRB; some batches differ.

Fix: Use the colour test (section 9) to find which order makes “red” actually red, and enter that value in `ws_pixel_order`. GRB is correct most of the time.

► **LED drifts to white when idle, “off” does not stay off (with SPI)**

Cause: The Pi leaves the SPI data line (MOSI) High between packets. The WS2812B reads this as all ones = white. Bright colours survive it, dim green and “off” flip.

Fix: Switch to the **PWM method (GPIO18)** — it pulls the line cleanly to Low. Move the data line to pin 12, set `led_type="ws2812"`, disable audio, run the service as root (sections 6–10).

► **Only one of several LEDs shows the status, the rest stay (white)**

Cause: `ws_count` is too low — surplus LEDs keep their last value.

Fix: Set `ws_count` to the actual number of LEDs and restart the service.

► **PWM: LED flickers or shows wrong colours despite correct wiring**

Cause: The onboard audio is still active and occupies the PWM hardware.

Fix: Enter `dtparam=audio=off` in `/boot/firmware/config.txt` and reboot (section 7).

► **OLED stays dark**

Cause: I2C not enabled, wrong address, PIL missing or wiring.

Fix: `sudo i2cdetect -y 1` — does 0x3c appear? Otherwise enable I2C, install `python3-pil/pillow`, check SDA/SCL/3V3/GND; if needed put 0x3d in `oled_addr`.

► **Neither LED nor OLED respond; `/dev/i2c-1` or `/dev/spidev0.0` are missing**

Cause: Not a real Raspberry Pi (e.g. LXC container/VM on x86) or the bus is not enabled.

Fix: `cat /proc/device-tree/model` must show a Pi. In an LXC/VM without a passed-through GPIO/I2C/SPI interface there are no pins — then run it on real Pi hardware.

► **After a system update the LED suddenly flickers again (PWM)**

Cause: An update reset `/boot/firmware/config.txt`; `dtparam=audio=off` is missing again.

Fix: Add the line again and reboot.

Useful diagnostic commands

```
cat /proc/device-tree/model           # real Pi?
ls -l /dev/i2c-1 /dev/spidev0.0      # buses present?
sudo i2cdetect -y 1                  # OLED at 0x3c?
groups YOUR_USER                     # in groups spi/i2c?
python3 /usr/local/bin/status_led.py # (stop the service) errors in foreground
journalctl -u status-led -e           # service log with errors
```